Angular-Similarity-Preserving Binary Signatures for Linear Subspaces

Jianqiu Ji, Jianmin Li, Qi Tian, *Senior Member, IEEE*, Shuicheng Yan, *Senior Member, IEEE*, and Bo Zhang

***Abstract*— We propose a similarity-preserving binary signature method for linear subspaces. In computer vision and pattern recognition, linear subspace is a very important representation for many kinds of data, such as face images, action and gesture videos, and so on. When there is a large amount of subspace data and the ambient dimension is high, the cost of computing the pairwise similarity between the subspaces would be high and it requires a large storage space for storing the subspaces. In this paper, we first define the angular similarity and angular distance between the subspaces. Then, based on this similarity definition, we develop a similarity-preserving binary signature method for linear subspaces, which transforms a linear subspace into a compact binary signature, and the Hamming distance between two signatures provides an unbiased estimate of the angular similarity between the two subspaces. We also provide a lower bound of the signature length sufficient to guarantee uniform distance-preservation between every pair of subspaces in a set. Experiments on face recognition, gesture recognition, and action recognition verify the effectiveness of the proposed method.**

***Index Terms*— Binary signature, angular similarity, locality-sensitive hashing.**

I. Introduction

IN COMPUTER vision and pattern recognition applications, linear subspace is an important representation that may better capture the structures of many kinds of data, e.g. objects under different lighting conditions (such as face images [1]) or viewpoints [2], motion videos [3]–[5], speeches [4], and so on. In the following, we use the term subspace for short. Based on this representation, many pattern recognition, machine learning and data mining algorithms can be built to accomplish tasks like clustering, approximate nearest neighbor search and so on [6]. Once subspace is used as a representation, the first key problem is to define a proper pairwise similarity or distance between subspaces. Furthermore, in large-scale problems, storing and processing large amounts of subspaces with high ambient dimensions require expensive storage and computational cost. A possible solution would be representing subspaces with short signatures that preserve pairwise similarity.

In this paper, first we define the concept of angle, pairwise angular similarity and angular distance between subspaces, and show that the angular distance is a metric that satisfies non-negativity, identity of indiscernibles, symmetry and triangle inequality. Then we develop a method to produce similarity-preserving binary signatures for subspaces, the Hamming distance between which provides an unbiased estimate of the pairwise angular distance. We provide a lower bound on the length of the binary signatures that suffices to guarantee uniform distance-preservation within a set of subspaces, which is similar to the Johnson-Lindenstrauss Lemma on random projection. In our previous work [7], we use the proposed method to generate angular-similarity-preserving binary signatures for face images, and demonstrate the effectiveness and efficiency of the method by an experiment on face recognition. As an extension to our previous work [7], this work extends the method to data cube, or 3rd-order tensor, such as action and gesture video sequences. We show that this kind of data can be naturally represented by 3 subspaces and thus we can develop binary signatures for fast data cube similarity search. Besides face recognition, we also conduct experiments on gesture and action recognition, showing that using the binary signature would achieve orders-of-magnitude faster recognition speed with little accuracy loss, while the storage space for the binary signatures is significantly reduced. We also compare our proposed method with Grassmannian-based Hashing (GBH) in the experiments. Since the proposed method provides a flexible way of achieving proper trade-offs between accuracy, speed and storage space, it can be used in environment where computation and storage resources are limited, such as mobile phone.

To the best of our knowledge, this is the first work on developing similarity-preserving binary signature for subspaces, and also the first for 3rd-order tensor. The advantages of similarity-preserving binary signature are two-fold:

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J. Ji, J. Li, and B. Zhang are with the State Key Laboratory of Intelligent Technology and Systems, Tsinghua National Laboratory for Information Science and Technology, Department of Computer Science and Technology, Tsinghua University, Beijing 100084, China (e-mail: jijq10@mails.tsinghua.edu.cn; lijianmin@mail.tsinghua.edu.cn; dcszb@mail.tsinghua.edu.cn).

Q. Tian is with the Department of Computer Science, University of Texas at San Antonio, San Antonio, TX 78249-1644 USA (e-mail: qitian@cs.utsa.edu).

S. Yan is with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore 117576 (e-mail: eleyans@nus.edu.sg).

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the storage space is significantly reduced due to the use of compact binary signatures; on the other hand, pairwise angular similarity or distance can be efficiently estimated by computing the Hamming distance between the corresponding binary signatures, and Hamming distance can be computed very fast in modern CPUs [8]. When the ambient dimension is high and the number of subspaces is huge, the similarity-preserving binary signature would serve as a fundamental tool towards different kinds of large-scale applications, e.g. clustering, approximate nearest neighbor search and so on.

II. Related Work

In recent years, intensive attention and research efforts have been devoted to developing small codes for vectors [8]–[16], in purpose of database compression and fast approximate nearest neighbor search. To the best of our knowledge, little research has been done on developing small codes for subspaces.

Various definitions of distance and similarity have been proposed for subspaces [4], [17], [18]. The distance definition in [17] is not shown to be metric, and all of these previous similarities or distances are not shown to admit a possible way to generate similarity-preserving binary signatures.

The works most related to ours are the approximate nearest subspace search methods [4], [19], [20]. These methods provide ways of conducting approximate nearest neighbor search in terms of specific distances in a database of subspaces. However, their methods are designed specifically for approximate nearest neighbor search, and they do not provide a compact representation, or more specifically, binary representation for subspaces.

The proposed binary signature for subspaces in this work can be used in not only approximate nearest neighbor search, but also applications where only the pairwise similarity or distance is exploited, e.g. clustering, kernel methods. In other words, generating binary signatures for subspaces may be regarded as a general preprocessing step.

III. Pairwise Similarity Measurement for Linear Subspaces

A. Principal Angles Between Two Linear Subspaces

The definition of principal angles between subspaces is a fundamental tool in mathematics, statistics and related applications, e.g. data mining. The concept was first introduced by Jordan [21] in 1875. Principal angles reveal the relative position of two subspaces. In this work, they are the building blocks of our definition of the angle, angular similarity and angular distance between subspaces.

Formally, for two subspaces $\mathcal{P}$ and $\mathcal{Q}$ of $\mathcal{R}^d$, denote the dimension $dim(\mathcal{P})$ of $\mathcal{P}$ by $d_{\mathcal{P}}$ and the dimension $dim(\mathcal{Q})$ of $\mathcal{Q}$ by $d_{\mathcal{Q}}$. Assume that $d_{max} = d_{\mathcal{P}} \geq d_{\mathcal{Q}} = d_{min}$. Then the principal angles between them, $\theta_1, \theta_2, ..., \theta_{d_{min}} \in [0, \pi/2]$, are defined recursively as [22], [23]:

$$\cos(\theta_i) = \max_{u_i \in \mathcal{P}} \max_{v_i \in \mathcal{Q}} \frac{u_i^T v_i}{\|u_i\| \|v_i\|}$$

subject to

$$u_i^T u_j = 0 \quad \text{and} \quad v_i^T v_j = 0, \quad \text{for } j = 1, 2, ..., i-1$$

Principal angles between two subspaces can be computed via singular value decomposition [22], [23]. Assume that P is a $d \times d_{\mathcal{P}}$ matrix with orthonormal columns which form an orthonormal basis of the subspace $\mathcal{P}$, and Q is a $d \times d_{\mathcal{Q}}$ matrix with orthonormal columns which form an orthonormal basis of the subspace $\mathcal{Q}$. Then the cosine of each of the principal angles equals a singular value of $P^T Q$. Formally, assume that the reduced singular value decomposition (SVD) of $P^T Q$ is

$$Y^T P^T Q Z = diag(\sigma_1, \sigma_2, ..., \sigma_{d_{min}}), \tag{1}$$

where $1 \geq \sigma_1 \geq \sigma_2 \geq ... \geq \sigma_{d_{min}} \geq 0$ are the singular values of $P^T Q$ and $Y \in \mathcal{R}^{d_{\mathcal{P}} \times d_{\mathcal{Q}}}$, $Z \in \mathcal{R}^{d_{\mathcal{Q}} \times d_{\mathcal{Q}}}$ have orthonormal columns.

Then

$$\theta_i = \arccos(\sigma_i), \quad i = 1, 2, ..., d_{min}. \tag{2}$$

B. Angle, Angular Similarity and Angular Distance Between Subspaces

As mentioned before, principal angles depict the relative position of two subspaces. In this section we formally define the angle, pairwise similarity measurement and distance for linear subspaces, based on principal angles.

Definition 1: For two subspaces $\mathcal{P}$ and $\mathcal{Q}$ of $\mathcal{R}^d$, denote the dimension $dim(\mathcal{P})$ of $\mathcal{P}$ by $d_{\mathcal{P}}$ and the dimension $dim(\mathcal{Q})$ of $\mathcal{Q}$ by $d_{\mathcal{Q}}$. Assume that $d_{max} = d_{\mathcal{P}} \geq d_{\mathcal{Q}} = d_{min}$, and $\theta_1, \theta_2, ..., \theta_{d_{min}}$ are the principal angles between $\mathcal{P}$ and $\mathcal{Q}$. Then the angle between $\mathcal{P}$ and $\mathcal{Q}$ is defined as

$$\theta_{\mathcal{P},\mathcal{Q}} = \arccos \frac{\sum_{i=1}^{d_{min}} \cos^2 \theta_i}{\sqrt{d_{max}}\sqrt{d_{min}}}. \tag{3}$$

Their angular similarity is

$$\begin{aligned} sim(\mathcal{P}, \mathcal{Q}) &= 1 - \frac{\theta_{\mathcal{P},\mathcal{Q}}}{\pi} \\ &= 1 - \frac{1}{\pi} \arccos \frac{\sum_{i=1}^{d_{min}} \cos^2 \theta_i}{\sqrt{d_{max}}\sqrt{d_{min}}}. \end{aligned} \tag{4}$$

And their angular distance is

$$\begin{aligned} d(\mathcal{P}, \mathcal{Q}) &= 1 - sim(\mathcal{P}, \mathcal{Q}) \\ &= \frac{1}{\pi} \arccos \frac{\sum_{i=1}^{d_{min}} \cos^2 \theta_i}{\sqrt{d_{max}}\sqrt{d_{min}}}. \end{aligned} \tag{5}$$

Since principal angles are invariant to the choice of orthonormal basis, this definition of angle, angular similarity and angular distance is also invariant to the choice of orthonormal basis. It is easy to see that the angular distance function $d(\cdot, \cdot)$ satisfies the following three properties: non-negativity, identity of indiscernibles and symmetry. Formally, for any subspaces $\mathcal{O}$, $\mathcal{P}$,$\mathcal{Q}$ of $\mathcal{R}^d$:

Non-negativity: $d(\mathcal{P}, \mathcal{Q}) \geq 0$

Identity of Indiscernibles: $d(\mathcal{P}, \mathcal{Q}) = 0$ if and only if $\mathcal{P} = \mathcal{Q}$

Symmetry: $d(\mathcal{P}, \mathcal{Q}) = d(\mathcal{Q}, \mathcal{P})$

Besides, later we will show that $d(\cdot, \cdot)$ also satisfies triangle inequality as described below, and thus it is a valid metric.

Triangle Inequality: $d(\mathcal{O}, \mathcal{Q}) \leq d(\mathcal{O}, \mathcal{P}) + d(\mathcal{P}, \mathcal{Q})$

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Note that to compute the angle $\theta_{\mathcal{P},\mathcal{Q}}$ between two subspaces $\mathcal{P}$ and $\mathcal{Q}$, it is not necessary to perform the SVD of P^TQ. Since $\|P^TQ\|_F^2 = \sum_{i=1}^{d_{min}} \cos^2\theta_i$, we may first compute $\|P^TQ\|_F^2$, then scale it by $\frac{1}{\sqrt{d_{max}}\sqrt{d_{min}}}$, and take the arccos of the value.

Besides these nice properties, in the subsequent section we will show that it is possible to generate binary signatures for subspaces, and the normalized Hamming distance between the binary signatures is an unbiased estimator of the angular distance. The experiment results on face recognition show that the angular distance as a distance measurement is very discriminative.

IV. Similarity-Preserving Binary Signature for Linear Subspaces

In this section we develop a way of producing binary signatures for linear subspaces, and the Hamming distance between binary signatures preserves angular similarity. We will show that the normalized Hamming distance between the binary signatures is an unbiased estimator of the pairwise angular distance (Definition 1) between the corresponding subspaces. We also provide a lower bound for the length of the binary signatures to guarantee uniform distance-preservation within a set of subspaces. In addition, we also show that the angular distance as defined satisfies triangle inequality.

Assume that P is a $d \times d_{\mathcal{P}}$ matrix with orthonormal columns which form an orthonormal basis of the subspace $\mathcal{P}$, and Q is a $d \times d_{\mathcal{Q}}$ matrix with orthonormal columns which form an orthonormal basis of the subspace $\mathcal{Q}$. Then we may represent subspaces $\mathcal{P}$ and $\mathcal{Q}$ by their orthographic projection matrices PP^T and QQ^T respectively. To generate binary signatures for $\mathcal{P}$, we use an operator $g(\cdot)$ to transform PP^T into a vector, as introduced in [4]. For a $d \times d$ symmetric matrix Z, the operator rearranges the elements of the upper triangular part of Z together with the diagonal elements scaled by $\frac{1}{\sqrt{2}}$, into a vector. Formally, it is defined as:

Definition 2 [4]:

$$g(Z) = [\frac{z_{1,1}}{\sqrt{2}}, z_{1,2}, ..., z_{1,d}, \frac{z_{2,2}}{\sqrt{2}}, z_{2,3}, ..., \frac{z_{d,d}}{\sqrt{2}}]^T \in \mathcal{R}^{d'}$$

where $d' = d(d+1)/2$, for any $d \times d$ symmetric matrix Z.

We will show that the angle between the two vectors $g(PP^T)$ and $g(QQ^T)$ equals the angle $\theta_{\mathcal{P},\mathcal{Q}}$ between $\mathcal{P}$ and $\mathcal{Q}$. Formally, we have

Lemma 1: Assume that P is a $d \times d_{\mathcal{P}}$ matrix with orthonormal columns which form an orthonormal basis of the subspace $\mathcal{P} \subseteq \mathcal{R}^d$, and Q is a $d \times d_{\mathcal{Q}}$ matrix with orthonormal columns which form an orthonormal basis of the subspace $\mathcal{Q} \subseteq \mathcal{R}^d$. Then $\theta_{g(PP^T),g(QQ^T)} = \theta_{\mathcal{P},\mathcal{Q}}$.

Proof: Assume that $d_{max} = d_{\mathcal{P}} \geq d_{\mathcal{Q}} = d_{min}$ and $\theta_1, \theta_2, ..., \theta_{d_{min}}$ are the principal angles between $\mathcal{P}$, $\mathcal{Q}$. It is shown [4] that $\|PP^T - QQ^T\|_F^2 = d_{\mathcal{P}} + d_{\mathcal{Q}} - 2\sum_{i=1}^{d_{min}} \cos^2\theta_i = 2\|g(PP^T) - g(QQ^T)\|_2^2$.

Since $2\|g(PP^T)\|_2^2 = d_{\mathcal{P}}$, $2\|g(QQ^T)\|_2^2 = d_{\mathcal{Q}}$,

$$\begin{aligned} &2\|g(PP^T) - g(QQ^T)\|_2^2 \\ &\quad = d_{\mathcal{P}} + d_{\mathcal{Q}} - 2\sqrt{d_{\mathcal{P}}}\sqrt{d_{\mathcal{Q}}}\cos\theta_{g(PP^T),g(QQ^T)} \end{aligned} \tag{6}$$

Then $\theta_{g(PP^T),g(QQ^T)} = \arccos\frac{\sum_{i=1}^{d_{min}}\cos^2\theta_i}{\sqrt{d_{max}}\sqrt{d_{min}}} = \theta_{\mathcal{P},\mathcal{Q}}$ □

Therefore the problem of generating similarity-preserving binary signatures for subspaces reduces to that of generating angle-preserving binary signatures for vectors.

Sign-random-projection [9] is a probabilistic method for generating binary signatures for vectors. The function is a combination of sign function $sgn(\cdot)$ together with a random projection. Formally, a sign-random-projection function $h_v(\cdot) : \mathcal{R}^d \to \{0, 1\}$ is defined as

Definition 3 [9]:

$$h_v(a) = sgn(v^Ta)$$

for any vector $a \in \mathcal{R}^d$, where v is a $d \times 1$ random vector, each entry of which is an independent standard normal random variable, and $sgn(\cdot)$ is defined as:

$$sgn(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

It is proven that [24], for any two given vectors a and b:

$$Pr[h_v(a) = h_v(b)] = 1 - \frac{\theta_{a,b}}{\pi} \tag{7}$$

where $\theta_{a,b} = \arccos(\frac{a^Tb}{\|a\|_2\|b\|_2})$, and $\|\cdot\|_2$ denotes the ℓ_2-norm of a vector.

Then the function for generating binary signatures of only 1 bit for subspaces is a combination of the sign-random-projection function $h_v(\cdot)$ together with the operator $g(\cdot)$ as defined in Definition 2, i.e. $h_v(g(\cdot))$ (note that here v is a $d' \times 1$ vector). To be concise, we define

Definition 4: $h_v(\mathcal{P}) = h_v(g(PP^T))$.

Formally, by Lemma 1 and Equation (7), we have

$$\begin{aligned} &Pr[h_v(\mathcal{P}) = h_v(\mathcal{Q})] \\ &\quad = Pr[h_v(g(PP^T)) = h_v(g(QQ^T))] \\ &\quad = 1 - \frac{\theta_{g(PP^T),g(QQ^T)}}{\pi} \\ &\quad = 1 - \frac{\theta_{\mathcal{P},\mathcal{Q}}}{\pi} \\ &\quad = sim(\mathcal{P}, \mathcal{Q}) \end{aligned} \tag{8}$$

Thus the function family $h_v(g(\cdot))$ is a locality-sensitive hash family for subspaces, and by the following lemma proven in [9], we have that the angular distance $d(\cdot,\cdot)$ defined in Definition 1 satisfies **triangle inequality**.

Lemma 2 [9]: For any similarity function $sim(\cdot,\cdot)$ that admits a locality-sensitive hash family $h \in \mathcal{F}$ operating on domain $\mathcal{D}$ that satisfies:

$$Pr[h(a) = h(b)] = sim(a, b), \tag{9}$$

for any a, $b \in \mathcal{D}$. Then $1 - sim(\cdot,\cdot)$ satisfies triangle inequality.

To produce binary signatures of length K for subspaces, we need to generate K standard normal random vectors $v_1, v_2, ...v_K$ and form a $K \times d'$ random projection matrix A with each v_i^T as one row, and thus each entry of A is an independently sampled standard normal variable. Therefore the final function is

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Definition 5:

$$h_A(\mathcal{P}) = sgn(A\ g(PP^T)) = [h_{v_1}(\mathcal{P}), h_{v_2}(\mathcal{P}), ..., h_{v_K}(\mathcal{P})]^T \quad (10)$$

where $A = [v_1, v_2, ...v_K]^T$, *and the sign function* $sgn(\cdot)$ *is applied element-wisely to its input.*

It can be shown that the normalized Hamming distance between two binary signatures $h_A(\mathcal{P})$ and $h_A(\mathcal{Q})$ is an unbiased estimator of the angular distance between the two subspaces $\mathcal{P}$ and $\mathcal{Q}$. Formally, by Equation (8) and the linearity of expectation, it can be shown that

$$\mathbb{E}[d_H(h_A(\mathcal{P}), h_A(\mathcal{Q}))/K] = \frac{\theta_{\mathcal{P},\mathcal{Q}}}{\pi} = d(\mathcal{P}, \mathcal{Q}) \quad (11)$$

where $d_H(\cdot,\cdot)$ outputs the Hamming distance between two binary vectors. Therefore the binary signature for subspaces is similarity-preserving.

The variance of this estimator is

$$Var[d_H(h_A(\mathcal{P}), h_A(\mathcal{Q}))/K] = \frac{\theta_{\mathcal{P},\mathcal{Q}}}{K\pi}(1 - \frac{\theta_{\mathcal{P},\mathcal{Q}}}{\pi}) \quad (12)$$

This shows that the longer the length of the binary signatures, the smaller the variance of the estimator.

The following lemma gives a lower bound of K to ensure that with constant probability, the binary signatures preserve the angular distance between every pair of the subspaces in a fixed set of n subspaces within a constant small error ϵ.

Lemma 3: For any $0 < \epsilon < 1, 0 < \delta < \frac{1}{2}$ *and any set* S *of* n *subspaces of* $\mathcal{R}^d$, *let* $K \geq K_0 = \frac{1}{2\epsilon^2}\ln\frac{n(n-1)}{\delta}$ *and* $h_A(\cdot)$ *as defined in Definition 5, and each entry of* A *is an independent standard normal variable. Then with probability at least* $1-\delta$, *for all* $\mathcal{P}, \mathcal{Q} \in S$:

$$|d_H(h_A(\mathcal{P}), h_A(\mathcal{Q}))/K - d(\mathcal{P}, \mathcal{Q})| \leq \epsilon \quad (13)$$

Proof: Denote $A = [v_1, v_2, ...v_K]^T$. For a fixed pair of subspaces $\mathcal{P}_0, \mathcal{Q}_0 \in S$. Define random variables

$$X_i = \begin{cases} 1, & h_{v_i}(\mathcal{P}_0) \neq h_{v_i}(\mathcal{Q}_0) \\ 0, & h_{v_i}(\mathcal{P}_0) = h_{v_i}(\mathcal{Q}_0) \end{cases}$$

for $i = 1, 2, ..., K$. Then $d_H(h_A(\mathcal{P}_0), h_A(\mathcal{Q}_0)) = \sum_{i=1}^K X_i$. By Hoeffding's inequality,

$$\begin{aligned} &Pr[|d_H(h_A(\mathcal{P}_0), h_A(\mathcal{Q}_0))/K - d(\mathcal{P}_0, \mathcal{Q}_0)| \geq \epsilon] \\ &= Pr[|\frac{1}{K}\sum_{i=1}^K X_i - \mathbb{E}(\frac{1}{K}\sum_{i=1}^K X_i)| \geq \epsilon] \\ &\leq 2\exp(-2\epsilon^2 K) \end{aligned} \quad (14)$$

Denote event $E = \{|d_H(h_A(\mathcal{P}), h_A(\mathcal{Q}))/K - d(\mathcal{P}, \mathcal{Q})| \leq \epsilon, for\ all\ \mathcal{P}, \mathcal{Q} \in S\}$. Then by union bound,

$$Pr[E] \geq 1 - \binom{n}{2}2\exp(-2\epsilon^2 K) \geq 1 - \delta \quad (15)$$

□

Remarks: Lemma 3 can generalize to any locality-sensitive hashing method that satisfies Equation (9). It shows that to guarantee the uniform distance-preservation within a set, the signature length K has nothing to do with the original ambient dimension of the input space, and it increases very slowly as the size of the set grows. This is similar to the Johnson-Lindenstrauss Lemma [25], [26], a well-known theoretical result on random projection.

V. Experiments

To demonstrate the effectiveness of the proposed binary signature as well as to showcase some potential applications, we conduct three sets of experiments: face recognition, gesture recognition and action recognition.

A. Face Recognition

In this section we demonstrate the effectiveness of the binary signature by applying it to face recognition.

Datasets: We use the union of two face datasets, the Extended Yale Face Database B [27], [28] and the PIE database [29]. Note that both of these datasets are the processed versions[1] [30]–[33]. Each image in both datasets has been cropped and resized to 32×32. Extended Yale Face Database B contains 38 individuals each with about 64 near-frontal images under different illuminations. PIE database contains 68 individuals each with 170 images under 5 near frontal poses (C05, C07, C09, C27, C29), 43 different illumination conditions and 4 different expressions. We only use the frontal face images (C27) from PIE database. Therefore the union of these two datasets contains 106 individuals in total. The two datasets have been split into training sets and testing sets respectively. The training set of the Extended Yale Face Database B contains around 50 images per individual, and the rest (around 14) images are in the testing set. The training set of the PIE database has about 20 images per individual, and the rest (around 14) images are in the testing set.

Experiment Setup: Each face image is vectorized to a 1024×1 vector. Several vectors of the same individual constitute a subspace. For each of these subspaces, we fit a subspace of dimension 9 by taking the first 9 principal components, and use this subspace to represent each individual in the training set. The same goes for the testing set, except that we test query subspaces with dimensions $dq = 4, 9, 13$ respectively. Thus there are 106 subspaces in the training set and testing set respectively.

The baseline is the exact nearest subspace search method in terms of angular distance. For each individual in the testing set (querying set), we search for the nearest subspace in the training set (database) in terms of angular distance as defined in Definition 1, and use its label as the prediction to that of the query. This is done by linear scanning the training set and returning the candidate subspace with the nearest angular distance to the query. Another method to compare is Grassmannian Based Hashing (GBH) [19], which uses specifically designed hash functions to transform subspaces into hash keys, and builds hash tables to enable fast search. In the experiment, we set $K = 7$, $S = 20$, $\theta = 1.475$ for GBH.

To test the effectiveness of the proposed binary signature, we generate a binary signature for each subspace in the training and the testing set, and thus each subspace is now represented by a binary signature. For each query we search

[1] http://www.cad.zju.edu.cn/home/dengcai/Data/FaceData.html

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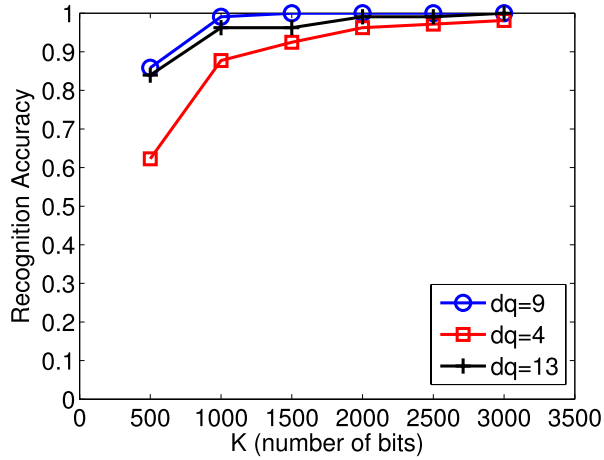

Fig. 1. Face recognition accuracy achieved by using the binary signatures of various lengths K (number of bits) on Extended Yale database B + PIE database, with query subspaces of dimensions $dq = 4, 9, 13$. The exact nearest subspace search in terms of angular distance achieves 100% accuracy for $dq = 4, 9, 13$.

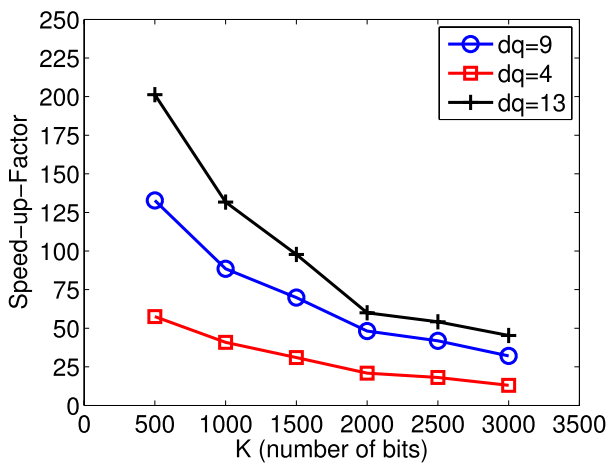

Fig. 2. Speed-up-factor achieved by using the binary signatures of various lengths K (number of bits) on Extended Yale database B + PIE database, with query subspaces of dimensions $dq = 4, 9, 13$.

for the nearest binary signature in the training set in terms of Hamming distance. This is also done by linear scanning the database. Since the length of the binary signatures is relatively small and Hamming distance can be computed very efficiently, linear scan in Hamming space is very fast in practice [8]. We test different lengths K of the binary signatures ranging from 500 to 3000.

Results: The accuracy of face recognition using the binary signatures with different lengths (number of bits) are shown in Figure 1. Note that the exact nearest subspace search in terms of angular distance achieves 100% accuracy for $dq = 4, 9, 13$. Under different query subspace dimensions, as the length K of the binary signatures grows, the recognition accuracy gradually increases and converges to 1. This is due to the asymptotic convergence guarantee of the estimators produced by the binary signatures. The accuracy of $dq = 9$ is the highest for various lengths K, which is slightly higher than that of $dq = 13$ for some values of K. Figure 2 shows

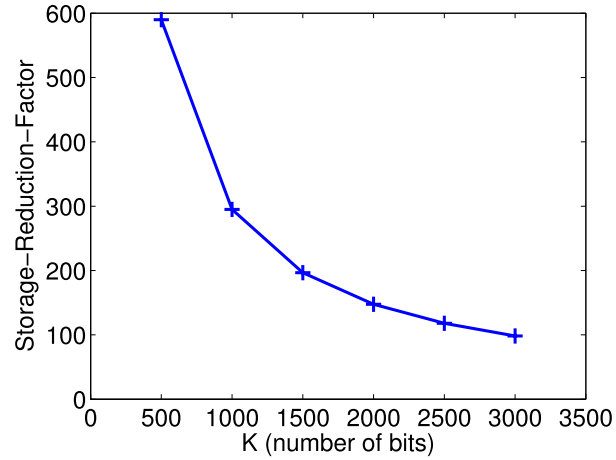

Fig. 3. Storage-reduction-factor achieved by using the binary signatures of various lengths K (number of bits) on Extended Yale database B + PIE database.

the factor of speed-up over the exact method achieved by using the binary signatures, against different lengths K, for $dq = 4, 9, 13$. As the query subspace dimension grows, the speed-up-factor increases accordingly. This is because when fixing the dimension of the subspaces in the database, the computational cost of computing the angular distance grows approximately linearly with the query subspace dimension: the computational cost of computing the angular distance between two subspaces $\mathcal{P}$ and $\mathcal{Q}$ is $O(d \times d_{\mathcal{P}} \times d_{\mathcal{Q}})$.

When $dq = 9$ and $K = 1500$, the approximate method using the binary signatures achieves more than 60 times faster recognition speed than exact method, with 100% recognition accuracy. If each entry of a vector is stored with 32 bits, then a subspace $\mathcal{P}$ of $\mathcal{R}^d$ with dimension $d_{\mathcal{P}}$ requires $32 \times d \times d_{\mathcal{P}}$ to store (the $d \times d_{\mathcal{P}}$ matrix P of which the columns form an orthonormal basis of $\mathcal{P}$). In the experiment, $d = 1024$, $d_{\mathcal{P}} = 9$, then the storage requirement for each subspace in the database is 294912 bits. Therefore using the binary signatures of 1500 bits for each subspace reduces the storage by a factor of 196. Figure 3 shows the storage-reduction-factor achieved by using the binary signatures of various lengths K.

Grassmannian Based Hashing achieves 83.02% accuracy, while it is only about 4 times faster than exact search, and its storage space is more than the exact search method, since all original data are stored. Note that in this experiment $dq = 9$.

B. Gesture and Action Recognition

Cambridge Gestures dataset contains 900 videos of hand gestures. There are three kinds of hand shapes: flat, spread, v-shape. And there are three kinds of motions (left, right, contraction). The combination of hand shapes and motions yield 9 classes of gestures. Each class contains 100 samples, evenly divided into 5 sets with 20 samples each. We use 1 set for testing and 4 sets for training. For each video sample, we select the middle 32 frames with each frame transformed to gray-scale and rescaled to 20×20.

Experiment Setup: Each frame sequence forms a 3D data cube, or a 3rd-order tensor, which can be

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unfolded along any of the three axes (x, y, t) to form three matrices respectively. For example, to unfold a data cube of a sequence of T frames to a matrix along axis t, we vectorize each frame to a column vector and constitute a matrix. Then the data cube can be represented by the subspace spanned by the columns of the matrix. To better exploit the subspace structure of a data cube, we also unfold it along x and y axes to two other matrices and use the two corresponding subspaces spanned by their columns respectively, to represent the data cube. Unfolding a data cube along three axes to make three subspaces is proposed in [20]. Then the dissimilarity between two data cubes can be defined as the sum of the three angles between the three pairs of corresponding subspaces.

Then for a data cube, we use three binary signatures to represent the corresponding three subspaces, and sum up the normalized Hamming distances between each pair of corresponding signatures of two data cubes to approximate the dissimilarity between the two data cubes.

In the experiment, the three signatures of a data cube are set to be the same length, and thus we can concatenate them to form a single signature, since the Hamming distance between two of these signatures approximates the dissimilarity, up to a constant scale factor. In this way, we generate a binary signature for each data cube in the training and the testing set. For each query we search for the nearest binary signature in the training set in terms of Hamming distance. To accelerate the signature generation process, we use a bilinear random projection technique similar to the bilinear random rotation method as described in [34]. We test different lengths K ranging from 600 to 7500.

The baseline method is exact nearest data cube search, in terms of the dissimilarity described above. The search scheme is linear scan through the database.

We also extend the GBH method to data cube search. For each of the three subspaces induced by a query data cube, we use GBH to search for a similar subspace in the database. The data cubes corresponding to the three subspaces returned by the GBH algorithm constitute the final candidate set. Finally we choose from this set the nearest data cube to the query. The distance between data cubes is the sum of the three distances between corresponding subspace pairs, and the distance between subspaces is defined as in GBH [19]. In the experiment, we set $K = 9$, $S = 20$, $\theta = 1.29$ for GBH.

Results: The accuracy of gesture recognition using the binary signatures with different lengths K (number of bits) are shown in Figure 4. The baseline method (exact nearest data cube search) achieves 96.11% accuracy. As the length K of the binary signatures grows, the recognition accuracy gradually increases and converges to that of the exact method. Figure 5 shows the factor of speed-up over the exact method achieved by using the binary signatures, against different lengths K. Figure 6 shows the storage-reduction-factor achieved by using the binary signatures of various lengths K.

When $K = 6000$, the binary signature method achieves about 200 times faster recognition speed than the exact method, with 93.94% recognition accuracy. And the storage space is reduced by a factor of more than 60, if each entry of a data cube is stored with 32 bits.

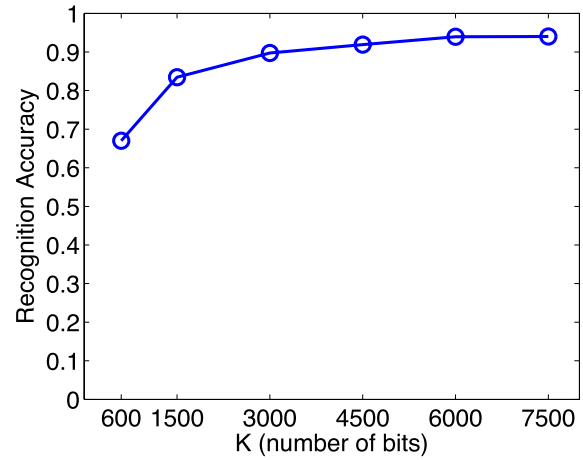

Fig. 4. Gesture recognition accuracy achieved by using the binary signatures of various lengths K (number of bits) on Cambridge Gestures dataset. The exact nearest subspace search achieves 96.11% accuracy.

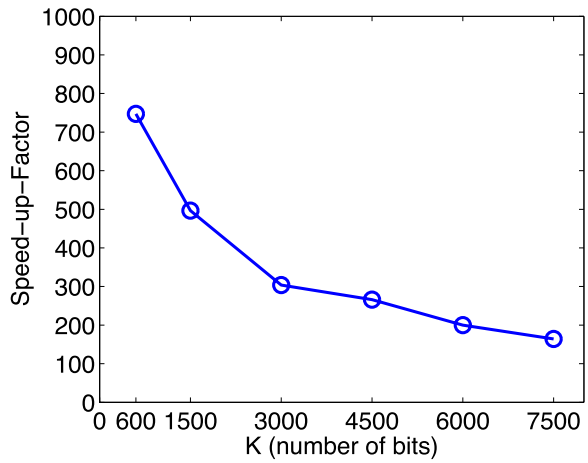

Fig. 5. Speed-up-factor achieved by using the binary signatures of various lengths K (number of bits) on Cambridge Gestures dataset.

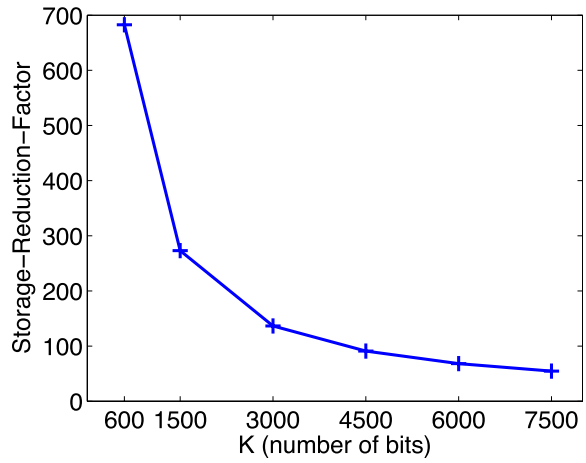

Fig. 6. Storage-reduction-factor achieved by using the binary signatures of various lengths K (number of bits) on Cambridge Gestures dataset.

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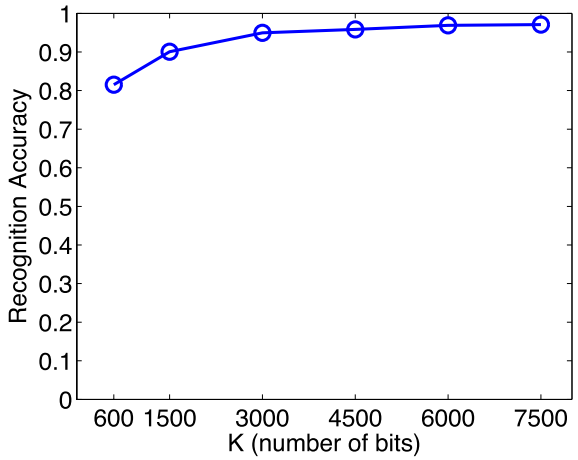

Fig. 7. Action recognition accuracy achieved by using the binary signatures of various lengths K (number of bits) on KTH dataset. The exact nearest subspace search in terms of angular distance achieves 98.15% accuracy.

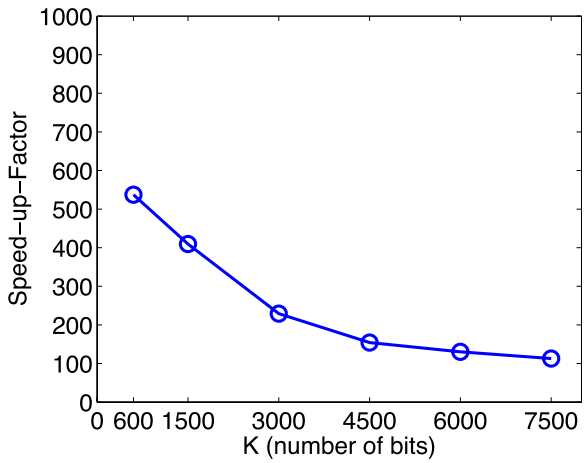

Fig. 8. Speed-up-factor achieved by using the binary signatures of various lengths K (number of bits) on KTH dataset.

Note that the recognition accuracy achieved by GBH is 88.33%, while the speed-up-factor of GBH is no more than 2.

KTH Action Dataset: Contains 600 videos with 25 subjects each performs 6 actions (boxing, clapping, jogging, waving, running, walking) in 4 scenarios. We crop the frames according to the annotations provided by [35], and transform them into gray-scale images and rescale them to 20×20. We take 24 consecutive frames for the temporal window. The dataset is divided into training set and testing set. The training set contains all the actions performed by 16 subjects, while testing set contains all that performed by the other 9 subjects.

The experiment setup is similar to gesture recognition.

Results: The accuracy of action recognition using the binary signatures with different lengths K (number of bits) are shown in Figure 7. The exact nearest data cube search, achieves 98.15% accuracy. As the length K of the binary signatures grows, the recognition accuracy gradually increases and converges to that of the exact method. Figure 8 shows the factor of speed-up over the exact method achieved by using the binary signatures, against different lengths K. Figure 9 shows the storage-reduction-factor achieved by using the binary signatures of various lengths K.

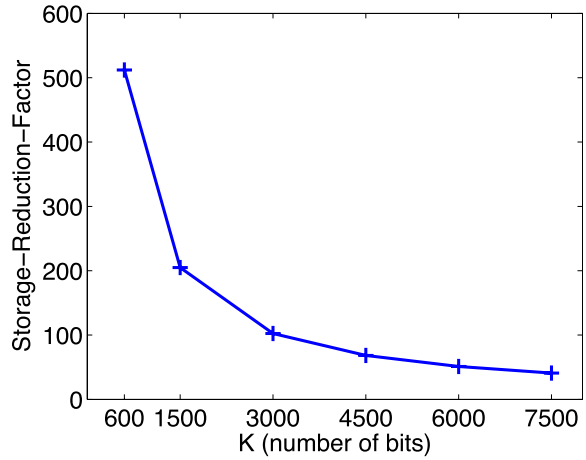

Fig. 9. Storage-reduction-factor achieved by using the binary signatures of various lengths K (number of bits) on KTH dataset.

When $K = 4500$, the binary signature method achieves more than 150 times faster recognition speed than the exact method, with 95.83% recognition accuracy. And the storage space is reduced by a factor of more than 60, if each entry of a data cube is stored with 32 bits.

The performance of GBH is similar as in the gesture recognition experiment. The recognition accuracy of GBH is 87.96%, while the speed-up-factor is no more than 2.

VI. Conclusion

In this paper we formally define the angular similarity and angular distance between subspaces, and we show that the angular distance is a metric which satisfies non-negativity, identity of indiscernibles, symmetry and triangle inequality. Then we propose a method to produce compact binary signatures for subspaces. The normalized Hamming distance between the binary signatures is an unbiased estimator of the pairwise angular distance. We provide a lower bound on the length of the signatures which guarantees uniform angular-distance-preservation within a set of subspaces. We also show that the binary signature can also be extended to data cube, or 3rd-order tensor, such as action and gesture videos. The experiments on face recognition, gesture and action recognition show that using the binary signature as representation achieves orders-of-magnitude speed-up, and a significant reduction of storage space, with little loss in recognition accuracy.

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Jianqiu Ji received the B.E. and Ph.D. degrees from the Department of Computer Science and Technology, Tsinghua University, in 2010 and 2015, respectively. His supervisor is Prof. B. Zhang. His current research interest is hashing method.

Jianmin Li received the Ph.D. degree in computer application from the Department of Computer Science and Technology, Tsinghua University, in 2003. He is currently an Associate Professor with the Department of Computer Science and Technology, Tsinghua University. He has authored over 50 journal and conference papers. His main research interests include image and video analysis, image and video retrieval, and machine learning. He received the second class Technology Innovation Award by the State Administration of Radio Film and Television in 2009.

Qi Tian (M'96–SM'03) received the B.E. degree in electronic engineering from Tsinghua University, China, in 1992, the M.S. degree in electrical and computer engineering from Drexel University, in 1996, and the Ph.D. degree in electrical and computer engineering from the University of Illinois at Urbana–Champaign, in 2002. He is currently a Professor with the Department of Computer Science, The University of Texas at San Antonio. He took a one-year faculty leave with Microsoft Research Asia from 2008 to 2009. His research interests include multimedia information retrieval and computer vision.

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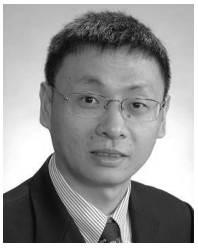

Shuicheng Yan is currently an Associate Professor with the National University of Singapore, and the Founding Lead of the Learning and Vision Research Group. His research areas include machine learning, computer vision, and multimedia, and he has authored or co-authored over 400 technical papers over a wide range of research topics, with Google Scholar citation over 17 000 times, and H-index of 57. He was an ISI Highly-Cited Researcher in 2014 and an IAPR Fellow in 2014. He received the Best Paper/Demo Award from ACM MM'13 (Best Paper and Best Student Paper), ACM MM12 (Best Demo), PCM'11, ACM MM10, ICME10, and ICIMCS'09, the runner-up prize of ILSVRC'13, the winner prize of ILSVRC14 detection task, the winner prize of the classification task in PASCAL VOC 2010-2012, the winner prize of the segmentation task in PASCAL VOC 2012, the honourable mention prize of the detection task in PASCAL VOC'10, the 2010 TCSVT Best Associate Editor (BAE) Award, the 2010 Young Faculty Research Award, the 2011 Singapore Young Scientist Award, and the 2012 NUS Young Researcher Award. He was/is Area Chair of CVPR, IJCAI, ACM MM, and among others, and the TPC Chair of ACM MM15, ACM MM 2017, and among others.

Bo Zhang received the B.E. degree from the Department of Automatic Control, Tsinghua University, China, in 1958. From 1980 to 1982, he visited the University of Illinois at Urbana–Champaign, USA, as a Scholar. He is currently a Professor with the Department of Computer Science and Technology, Tsinghua University. His research interests include artificial intelligence, machine learning, pattern recognition, knowledge engineering, intelligent robotics, and intelligent control. He is a member of the Chinese Academy of Sciences. He won the Prize of European Artificial Intelligence in 1984. He won the first class of Science and Technology Progress Prize three times in 1987, 1993, and 1998, and the third class Prize two times in 1995 and 1999, respectively.

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